Anika Singhania

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EDUCATION

B.Tech in Information Technology

College of Engineering, Guindy
CGPA: 8.70/10.0 (2022-2026)

SKILLS

Languages
Go, Java, Python, TypeScript, C++, SQL

Frameworks & Systems
Spring Boot, Node.js, FastAPI, React, Next.js, gRPC

Databases & Storage
PostgreSQL, Redis, MongoDB, Apache Kafka, Elasticsearch

Cloud & DevOps
Docker, Kubernetes, AWS (EC2, S3, RDS), Git, CI/CD Pipelines

EXPERIENCE

Platform Engineer — Groww (Nextbillion Technology) *Jun 2025 – Present*

- Designed real-time event pipeline consuming 5M daily telemetry records via Apache Kafka, providing sub-second fraud detection alerts.
- Optimized database queries and added composite b-tree indexes on partitioned ledger tables, lowering average query execution time from 420ms to 35ms.
- Engineered distributed idempotency layer with Redis and PostgreSQL, eliminating 100% of double-debit anomalies during peak flash sales.

PROJECTS

High-Concurrency Real-Time Collaboration Canvas [TypeScript, WebSocket, CRDT, React, Node.js]

- Built multi-user collaborative workspace using Conflict-free Replicated Data Types (CRDTs) to ensure convergence across disconnected edits.
- Engineered WebSocket server in Node.js with delta-compression protocol, reducing network bandwidth consumption by 64% during intense cursor movements.
- Authored comprehensive property-based test suite with fast-check verifying eventual consistency across simulated random network partitions.

Peer-to-Peer File Transfer Protocol with NAT Traversal [Rust, STUN/TURN, UDP, Tokio, Cryptography]

- Developed user-space P2P file synchronization tool implementing UDP hole-punching and ICE protocol to bypass restrictive symmetric NATs.
- Utilized ChaCha20-Poly1305 authenticated encryption to guarantee confidentiality and packet integrity over public Wi-Fi networks.
- Achieved wire-speed throughput of 940 Mbps over local gigabit connections using zero-copy memory-mapped file buffers.